

C635

MA, Mathematics Education (K-6) Capstone

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A second-grade teacher at a school with a new math coach is choosing a research question for her capstone. Her principal encourages teachers to present at the spring conference. Which question is best?

- A. How can I teach math better?
- B. What is the effect of daily number-talk routines on second graders' mental computation scores over eight weeks?
- C. Should every school adopt number talks?
- D. Is math important for young children?

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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2.

A teacher who has collected 25 articles on fraction manipulatives is drafting her literature review. She keeps her notes in a spiral notebook. How should the review be organized?

- A. One paragraph summarizing each article in the order she found them
- B. Only articles from the past year, since older research is always outdated
- C. By themes, comparing findings across studies and ending with the gap her study will fill
- D. Only the studies that support her expected result

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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3.

A researcher at a K-6 school with a large garden wants to know how teachers perceive the new district math curriculum. She has a quiet office for meetings. Which design fits the question best?

- A. An experiment with a randomly assigned control group
- B. A quantitative analysis of student test scores
- C. Any design, since all designs answer all questions equally
- D. A qualitative design using interviews or focus groups

Target word _____	The one fact that decides _____	Answer ___	☐ sure ☐ unsure ☐ guess
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4. [TRANSFER — reversed stem / different variable]

A researcher will compare end-of-unit scores between classes that used manipulatives and classes that did not. Which design is this?

- A.** Ethnography
- B.** Mixed methods, which is required for every capstone
- C.** Qualitative case study
- D.** Quantitative, quasi-experimental

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A third-grade teacher's class of 22 students averaged 58% before a multiplication-games unit and 81% after eight weeks. The class also had a new reading program that year. She has no comparison class. What can she claim?

- A.** A class of 22 is too small to report any results.
- B.** Scores improved, but other causes such as maturation cannot be ruled out.
- C.** The gain is meaningless and should not be reported.
- D.** The games caused the 23-point gain.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

Which change would most strengthen a researcher's claim that an intervention caused a gain?

- A.** Survey teachers about whether they liked the intervention
- B.** Extend the unit from eight to twelve weeks
- C.** Add a similar comparison class that does not receive the intervention
- D.** Give the same class more post-tests

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A fifth-grade teacher at a charter school with a waiting list studies her own class of 26 students and finds a strong gain. Her room has a view of the mountains. How far can she generalize?

- A. Mainly to her own class or very similar settings
- B. To all fifth graders in the state
- C. Not even to her own class, since the sample was not random
- D. To all fifth graders, because she used random assignment

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8.

A teacher who presented at a regional conference last year runs a t-test comparing two groups and gets $p = .03$ with alpha set at $.05$. The test was run on a laptop borrowed from the library. What should she conclude?

- A. The difference is statistically significant; it would be unlikely if there were no true difference.
- B. There is a 3% chance the intervention works.
- C. The result is not significant because $.03$ is below $.05$.
- D. The intervention is proven to work for all students.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9. [TRANSFER — reversed stem / different variable]

A study reports $p = .20$ with alpha = $.05$. What is the correct conclusion?

- A. The result is significant at the $.05$ level.
- B. Fail to reject the null; no statistically significant difference was found.
- C. It proves the intervention has no effect.
- D. There is an 80% chance the intervention works.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A teacher at a school with a four-day week gives the same 24 students a fluency test before and after a six-week intervention. She uses a spreadsheet her colleague built. Which test compares the two sets of scores?

- A. No test; just compare the two averages
- B. Independent-samples t-test
- C. Paired-samples t-test
- D. Chi-square test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A researcher compares post-test means of an intervention class and a different comparison class. Which test fits?

- A. Independent-samples t-test
- B. Correlation
- C. Paired-samples t-test
- D. Chi-square test

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A kindergarten teacher with a popular class blog plans to record students solving counting tasks for her study. The school just updated its playground. What must happen before she collects data?

- A. Collect the recordings first, then ask for permission.
- B. Nothing, since these are her own students.
- C. Post the videos on her class blog to share results.
- D. Get IRB and district approval, then parental consent and student assent, and plan to protect identities.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A second-grade teacher ~~at a school with a new math coach~~ is choosing a research question for her capstone. ~~Her principal encourages teachers to present at the spring conference.~~ Which question is best?

Target word: best research question **Decides:** Specific focus, population, outcome, and feasible data.

A	Dropped element	Drops a specific intervention, population, and measurable outcome.
B	Correct	Names the intervention, population, outcome, and time frame.
C	Absolute language	A policy question too broad for one classroom study.
D	Adjacent category	A general opinion question, not researchable with study data.

2. Answer C Concept 2

A teacher who has collected 25 articles on fraction manipulatives is drafting her literature review. ~~She keeps her notes in a spiral notebook.~~ How should the review be organized?

Target word: organization of the review **Decides:** Synthesis by theme with a gap.

A	Confusion set	That is an annotated bibliography, not a synthesis.
B	Absolute language	Foundational and older studies can be essential.
C	Correct	A review synthesizes rather than lists.
D	Dropped element	Drops contrary evidence the review must address.

3. Answer D Concept 3

A researcher at a K-6 school ~~with a large garden~~ wants to know how teachers perceive the new district math curriculum. ~~She has a quiet office for meetings.~~ Which design fits the question best?

Target word: design for a perception question **Decides:** "Perceive" signals qualitative.

A	Confusion set	Tests an effect, not how teachers perceive something.
B	Adjacent category	Scores do not capture teacher perceptions.
C	Absolute language	The question determines the design.
D	Correct	Perceptions and experiences are best explored qualitatively.

4. Answer D Concept 3 · transfer

A researcher will compare end-of-unit scores between classes that used manipulatives and classes that did not. Which design is this?

Target word: reverse: name the design from the plan **Decides:** Comparing measured scores between groups.

A	Confusion set	Ethnography studies culture through immersion.
B	Absolute language	Mixed methods is one option, used when both data types are needed.
C	Inverted operation	A case study explores a case in depth; this compares scores.
D	Correct	Measured outcomes compared across existing groups.

5. Answer B Concept 4

A third-grade teacher's class of 22 students averaged 58% before a multiplication-games unit and 81% after eight weeks. ~~The class also had a new reading program that year.~~ She has no comparison class. What can she claim?

Target word: supported claim **Decides:** No comparison group.

A	Camouflage	Class size is not the main issue; the missing comparison is.
B	Correct	A single-group design supports growth claims, not causal ones.
C	Inverted operation	The gain is real data; only the causal claim is limited.
D	Absolute language	Without a comparison group, causation is not established.

6. Answer C Concept 4 • transfer

Which change would most strengthen a researcher's claim that an intervention caused a gain?

Target word: reverse: strengthen causal inference **Decides:** Add a comparison.

A	Adjacent category	Opinions do not establish cause.
B	Confusion set	More time adds exposure but still no comparison.
C	Correct	Lets her separate the intervention from maturation and history.
D	Dropped element	More testing of one group does not rule out other causes.

7. Answer A Concept 5

A fifth-grade teacher at a charter school ~~with a waiting list~~ studies her own class of 26 students and finds a strong gain. ~~Her room has a view of the mountains.~~ How far can she generalize?

Target word: generalizability **Decides:** Convenience sample of one class.

A	Correct	A convenience sample limits generalization.
B	Absolute language	One class is not a representative sample.
C	Inverted operation	Results still describe the class she studied.
D	Confusion set	Studying one intact class is not random assignment.

8. Answer A Concept 6

A teacher ~~who presented at a regional conference last year~~ runs a t-test comparing two groups and gets $p = .03$ with alpha set at $.05$. ~~The test was run on a laptop borrowed from the library.~~ What should she conclude?

Target word: interpretation of $p = .03$ **Decides:** $p < \alpha$: reject the null.

A	Correct	p below alpha leads to rejecting the null hypothesis.
B	Confusion set	p is not the probability that the hypothesis is true.
C	Inverted operation	Below alpha is significant.
D	Absolute language	Significance does not prove or generalize to all.

9. Answer B Concept 6 · transfer

A study reports $p = .20$ with $\alpha = .05$. What is the correct conclusion?

Target word: interpretation of a non-significant p **Decides:** $p > \alpha$.

A	Inverted operation	.20 is above .05.
B	Correct	The data do not show a reliable difference.
C	Absolute language	Failing to find an effect is not proof of none.
D	Confusion set	p is not the probability of the hypothesis.

10. Answer C Concept 7

A teacher at a school ~~with a four-day week~~ gives the same 24 students a fluency test before and after a six-week intervention. ~~She uses a spreadsheet her colleague built.~~ Which test compares the two sets of scores?

Target word: test for pre/post same students **Decides:** Same participants measured twice.

A	Dropped element	Drops the check for whether the difference is beyond chance.
B	Confusion set	That is for two different groups.
C	Correct	Each student is compared with themselves.
D	Adjacent category	Chi-square is for categorical counts, not mean scores.

11. Answer A Concept 7 · transfer

A researcher compares post-test means of an intervention class and a different comparison class. Which test fits?

Target word: reverse: test for two groups **Decides:** Two different groups.

A	Correct	Different students in each group.
B	Confusion set	Measures association, not a difference between groups.
C	Inverted operation	Paired tests need the same participants measured twice.
D	Adjacent category	For counts in categories, not means.

12. Answer D Concept 8

A kindergarten teacher ~~with a popular class blog~~ plans to record students solving counting tasks for her study. ~~The school just updated its playground.~~ What must happen before she collects data?

Target word: ethical step before data collection **Decides:** Approval and consent first.

A	Inverted operation	Consent must come before data collection.
B	Absolute language	Research with one's own students still needs approval and consent.
C	Confusion set	Confuses sharing results with protecting confidentiality.
D	Correct	Ethics approvals and consent precede data collection.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	9	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	1	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Researchable questions	A good research question is specific, names the population, intervention or focus, and outcome, and can be answered with data the researcher can feasibly collect.	Asking a broad, value-laden, or yes/no opinion question.	"Which research question is best?"
2	Literature review as synthesis	A literature review organizes sources by theme, compares findings, and identifies the gap the study will address.	Writing one summary paragraph per article, or citing only supportive studies.	"How should the review be organized?"
3	Design fits the question	Questions about measured effects or differences call for quantitative designs; questions about experiences and perceptions call for qualitative designs; mixed methods combines both for a reason.	Choosing a design by preference rather than by the question.	Words like "perceive, experience" vs. "effect, difference, score".
4	Internal validity	Without a comparison group, a pre/post gain cannot rule out maturation, history, or testing effects, so causal claims must be limited.	Claiming a single-group gain proves the intervention caused it.	One class, pre and post, no comparison.
5	Sampling and generalizability	A convenience sample from one's own classroom limits how far results generalize; random sampling supports wider generalization.	Generalizing results from one class to all students.	"My students" and a claim about "all".
6	Interpreting p-values	A p-value below alpha (such as .05) means the result would be unlikely if there were no true effect, so the null is rejected; it does not prove the hypothesis or give the probability the intervention works.	Reading p as the probability the intervention works, or treating non-significance as proof of no effect.	A p-value and "what can the researcher conclude?"
7	Matching the statistical test	Use a paired-samples t-test for the same participants measured twice and an independent-samples t-test for two different groups; chi-square is for categorical counts.	Using an independent test on pre/post data from one group.	"Same students" vs. "two classes".
8	Research ethics with children	Before collecting data, obtain required approvals (IRB and district) plus parental consent and student assent, and protect confidentiality.	Assuming a teacher's own class needs no consent, or seeking permission after data collection.	A teacher researching her own students.

Course scope (public description): MA, Mathematics Education (K-6) Capstone Written Project takes the student through the steps of planning and conducting research on a topic or issue related to the students' practice setting. The result is expected to be a significant piece of research, culminating in a written research report,

including sections describing a literature review, methodology, and detailed analysis and reporting of results. Prerequisite Courses: Research Foundations (C224), Research Questions and Literature Review (C225), Research Design and Analysis (C226), and Research Proposals (C227) or permission of a faculty manager. Additionally, students wishing to add the Capstone with fewer than eight weeks remaining in the term must receive permission of the faculty manager.